SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
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Course code: 10214AD701 **Course Name: Major Project**

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2
3799	Dr.P.SANTHI	22271	K. VENKATESH	22706	R. NARESH

Title of the Project: Spatio-Temporal Traffic Prediction in Urban Area using Convolutional Temporal Network System

Problem Statement
With rapid urban growth, traffic congestion has become a serious problem in urban areas, affecting travel time, fuel consumption, and the environment. Existing traffic prediction methods often fail to capture both the spatial relationships among roads and the temporal variations in traffic flow. As a result, these systems provide less accurate predictions during peak hours, accidents, or sudden traffic changes. Therefore, there is a need for an intelligent model that can simultaneously learn spatial and temporal traffic patterns. The proposed Convolutional Temporal Network System improves the accuracy of spatio-temporal traffic prediction in urban areas.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	The project requires knowledge of traffic flow, urban road networks, deep learning, convolutional networks, and temporal modeling. It involves designing a prediction system that analyzes spatial and temporal traffic patterns from large-scale datasets. Recent research on intelligent transportation and traffic prediction is also

				needed to improve accuracy and performance.
WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	The project involves conflicting requirements such as prediction accuracy versus computational cost, real-time processing versus model complexity, and large-scale traffic data handling versus system efficiency. It must also balance reliable traffic forecasting with changing urban conditions such as peak hours, accidents, and road closures.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	The project has no direct or standard solution because traffic patterns vary across locations and time. It requires original analysis to model complex spatial and temporal relationships, select suitable deep learning techniques, and optimize the prediction system for accurate urban traffic forecasting.
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	It deals with novel and infrequently encountered problems in urban traffic prediction, where traffic conditions continuously change due to road usage, peak hours, accidents, and environmental factors. Integrating spatial and temporal learning in a single model requires advanced and non-routine engineering approaches.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	The project addresses traffic prediction problems that are not fully covered by existing engineering standards or conventional traffic management practices. Therefore, research-based methods and engineering judgment are required to

				design an effective spatio-temporal prediction system.
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	The project involves multiple stakeholders such as city traffic authorities, commuters, urban planners, and emergency services, each with different requirements. The system must balance accurate prediction, fast response, ease of use, and efficient traffic management to satisfy these diverse needs.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The project consists of interconnected components such as traffic data collection, preprocessing, spatial feature extraction, temporal modeling, and prediction output. These components depend on each other and require a complete systems approach for accurate traffic forecasting.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project uses diverse resources including large-scale traffic datasets, urban road network information, deep learning frameworks, computing systems, and human resources such as researchers, traffic authorities, and system users.
EA2	Level of Interactions	Optimization of complex interactions	The project requires effective interaction between traffic data processing, spatial and temporal modeling, prediction accuracy, and real-time system performance. It must optimize these interconnected factors to provide reliable traffic forecasts.
EA3	Innovation	Creative and research-based solutions	The project involves innovative use of convolutional and temporal network techniques to predict traffic conditions by learning both spatial and temporal patterns. It applies research-based methods to improve the accuracy and efficiency of urban traffic forecasting.
EA4	Consequences to Society	Significant societal impact	The project has significant societal and environmental impact because accurate traffic

EA No.	Attribute	Characteristics	Justification based on the Project
	and Environment		prediction can reduce congestion, travel time, fuel consumption, and air pollution. It also supports safer and more efficient urban transportation systems.
EA5	Familiarity	Extends beyond prior experience	The project extends beyond conventional traffic prediction methods by applying advanced deep learning and spatio-temporal analysis techniques. It requires principle-based approaches to solve complex and changing urban traffic problems.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	The project supports the development of intelligent transportation infrastructure by using advanced traffic prediction techniques to improve urban mobility, innovation, and smart city planning.
SDG 16	Peace, Justice and Strong Institutions	The project contributes to better urban governance by supporting traffic authorities with accurate and reliable traffic predictions, leading to improved decision-making, reduced congestion, and more efficient public services.
SDG 11	Sustainable Cities and Communities	The project supports sustainable urban development by improving traffic management, reducing congestion, and enabling efficient transportation systems for safer and smarter cities.
SDG 3	Good Health and Well-being	The project contributes to public health by reducing traffic congestion, fuel consumption, and air pollution, which can improve the overall well-being of people in urban areas.

Overall Justification: The project satisfies the characteristics of Complex Engineering Problems and Activities because it requires advanced knowledge of traffic systems, deep learning, and spatio-temporal modeling. It addresses conflicting requirements such as prediction accuracy, real-time performance, and large-scale data handling. The project also uses innovative and research-based methods to solve complex urban traffic problems and has significant social and environmental benefits through improved traffic management and sustainable urban transportation.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels**:

K3 – Engineering Fundamentals

- Understanding of traffic flow principles, urban transportation systems, and road network structures.
- Knowledge of data preprocessing and the fundamentals of machine learning and deep learning for traffic prediction.

K4 – Specialist Knowledge

- Understanding of spatio-temporal traffic analysis, large-scale traffic datasets, and intelligent transportation systems.
- Advanced knowledge of deep learning techniques such as convolutional networks and temporal sequence models.

K5 – Engineering Design

- Design of an integrated spatio-temporal traffic prediction architecture using convolutional and temporal network models.
- Development of data processing, feature extraction, and prediction modules for accurate urban traffic forecasting.

K6 – Engineering Practice

- Practical implementation of the traffic prediction system using deep learning frameworks and large-scale traffic datasets.
- Testing, performance evaluation, and optimization of the model for accurate and real-time traffic forecasting.

K7 – Engineering Comprehension

- Ability to analyze traffic behavior under different conditions such as peak hours, accidents, and road congestion.
- Interpretation of prediction results and system performance to support better traffic management decisions.

K8 – Research Literature and Emerging Technologies

- Study of recent research literature on intelligent transportation systems and spatio-temporal traffic prediction.
- Use of advanced and emerging deep learning methods to improve prediction accuracy and system performance.

Signature

Dr. P. Santhi